

AI-Assisted Financial Analysis & Anomaly Detection

A synthetic corporate expense-review demonstration using Python, Excel and AI-assisted financial reasoning

Executive Summary

This portfolio project demonstrates how a finance professional can combine domain knowledge, transparent data-analysis rules and generative AI support to improve the efficiency of financial review. The analysis uses 1,000 synthetically generated corporate expense transactions covering January to June 2026. The objective is to identify transactions and spending patterns that may warrant further investigation while retaining human validation and professional judgement.

1,000 Transactions	CNY 1.90M Total Expense	78 Flagged Transactions
28 High-Value Flags	28 Duplicate-Like Flags	22 Weekend Transactions

Important: Review flags indicate items requiring additional investigation; they do not constitute evidence of fraud, misconduct or accounting error. All data in this project are synthetically generated and contain no confidential client, employer or audit information.

Project Objectives

- Structure and clean a synthetic financial transaction dataset.
- Identify potentially unusual transactions using transparent review rules.
- Assess duplicate-like patterns, reimbursement frequency and vendor concentration.
- Summarise department-level spending and review priorities.
- Demonstrate an auditable Human-AI collaboration workflow.

1. Data & Methodology

Dataset Scope

Field	Project Scope
Period	1 January 2026 - 30 June 2026
Transactions	1,000
Departments	6
Vendors	34
Currency	CNY
Total expense	CNY 1,901,417
Average transaction	CNY 1,901.42
Median transaction	CNY 895.00
Largest transaction	CNY 32,056

Review Rules

Review Area	Rule / Rationale
High-value	Flag transactions above the 97th percentile within each expense category.
Duplicate-like	Match on date, employee, expense category, vendor and amount.
Weekend	Flag transactions recorded on Saturday or Sunday for contextual review.
High-frequency	Identify employee-days containing 5 or more transactions.
Vendor concentration	Rank vendors by total spend and share of total expenditure.
Department spike	Compare department-month spending against its own historical pattern.

Workflow

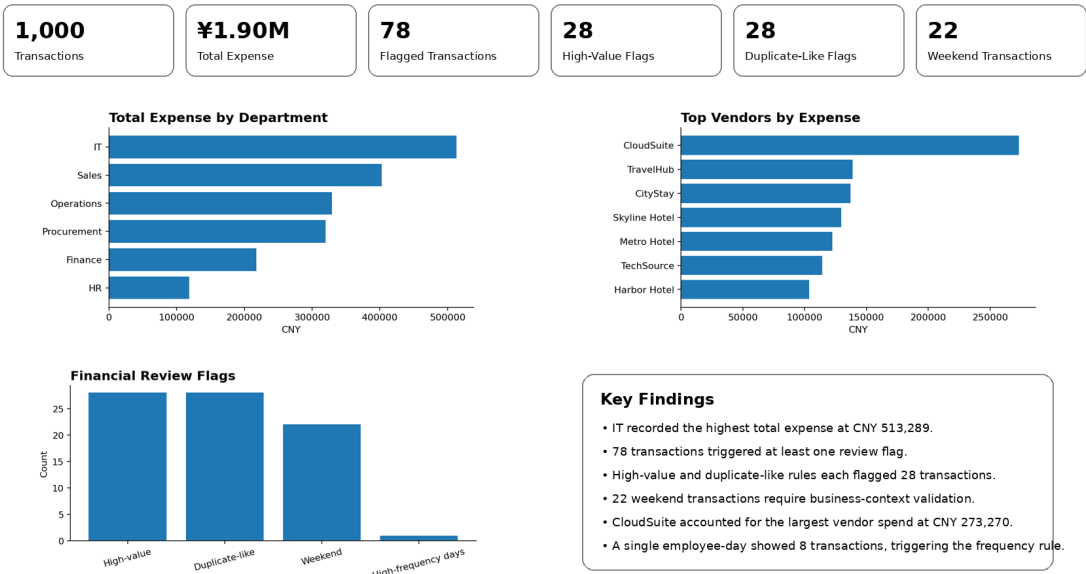
Business context -> Data cleaning -> Financial rule design -> Anomaly detection -> AI-assisted interpretation -> Human validation -> Risk interpretation -> Dashboard & findings

The methodology intentionally uses explainable, rule-based checks rather than a black-box model. This makes each review flag traceable to the source transaction and easier for a finance reviewer to challenge, validate and document.

2. Dashboard & Key Findings

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Synthetic corporate expense dataset | Python + Excel + AI-assisted financial review



Key Findings

- IT recorded the highest total expense at CNY 513,289, followed by Sales at CNY 402,943.
- 78 transactions triggered at least one transaction-level review flag.
- 28 high-value transactions were identified using category-specific percentile thresholds.
- 28 transactions belonged to duplicate-like groups based on matching date, employee, category, vendor and amount.
- 22 weekend transactions were identified for additional business-context validation.
- One employee-day contained 8 transactions, exceeding the high-frequency review threshold.
- CloudSuite was the largest vendor by total spend at CNY 273,270, approximately 14% of total expenditure.

Department Spending

Department	Transactions	Total Expense (CNY)	Average (CNY)
IT	172	513,289	2,984.24
Sales	220	402,943	1,831.56
Operations	205	329,163	1,605.67
Procurement	168	319,924	1,904.31
Finance	146	217,708	1,491.15
HR	89	118,390	1,330.22

3. Human-AI Collaboration & Controls

The project treats generative AI as an analytical assistant rather than a substitute for financial judgement. The central control principle is that AI can accelerate research, coding and interpretation, but final conclusions must remain traceable to the underlying data and validated by a human reviewer.

AI-Assisted Activities	Human Responsibilities
Support Python code development	Define business objectives and risk criteria
Suggest transparent anomaly-detection logic	Validate thresholds and review rules
Help structure financial analysis	Trace flags to source transactions
Summarise preliminary patterns	Assess economic substance and supporting evidence
Support interpretation and follow-up questions	Make final financial, control and escalation judgements

Validation Questions

- Is there a valid business reason for an unusually high amount?
- Does the transaction have complete supporting documentation?
- Are duplicate-like entries separate legitimate expenses or potential duplicate claims?
- Are weekend transactions consistent with travel or operational requirements?
- Does repeated spending with one vendor reflect an approved contract or a concentration risk?
- Is a departmental spending spike linked to a planned project, renewal or one-off investment?

Limitations

This is a synthetic demonstration rather than an audit or fraud-detection model. The rules are intentionally simple and explainable, and the dataset was designed to contain selected review patterns. In a real finance environment, thresholds would need to reflect company policy, materiality, business seasonality, approval limits, procurement arrangements and access to supporting documentation.

Conclusion

The project demonstrates a practical finance-analytics workflow in which AI and Python help focus human attention on potentially unusual transactions without replacing professional judgement. The combination of transparent rules, reproducible analysis, source-data traceability and human validation provides a more defensible approach to applying generative AI in finance, accounting and internal-control contexts.

Repository structure: synthetic dataset (Excel), reproducible Python notebook, dashboard image and this summary report.